COLLEGE OF BUSINESS EDUCATION

DAR ES SALAAM CAMPUS

COURSE: BACHELOR OF INFORMATION TECHNOLOGY

MODULE NAME: Multimedia Systems

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STREAM: TNC BIT - 2

INDIVIDUAL ASSIGNMENT

ElectroShop_Documentation

1. Project Overview

ElectroShop is a comprehensive, full-stack e-commerce platform designed for buying and selling electronics. It features a robust PHP backend, a responsive vanilla JavaScript frontend, and a normalized MySQL database. The system supports multiple user roles (Customer, Seller, Admin) with specific permissions and dashboards.

Key Features

Role-Based Access Control (RBAC):Distinct interfaces and capabilities for Admins, Sellers, and Customers.
Where By:

1. Admin

- **Responsibilities:**
    - Manage all users (promote/demote to Seller).
    - Add new stock to inventory.
    - Full CRUD on all tables.
    - Configure system settings.
    - View all reports and logs.

2. Seller

- **Responsibilities:**
    - Update user interface (post electronics).
    - Manage orders for clients.
    - Send feedback to clients.
    - See available stock.
    - CRUD only on their own products.
    - View/manage their own orders.
    - Limited reporting (sales, inventory).

3. Client (Customer)

- **Responsibilities:**
    - Read product catalog.
    - Create orders.
    - Update their own profile.
    - View their own order history.

Product Management:Full CRUD operations for products with categorization and search/filter capabilities.
Order Processing:Shopping cart functionality, order placement, and history tracking.
Inventory Management:Real-time stock tracking and inventory logs.
Security: secure password hashing , PDO prepared statements (SQL injection protection), and input sanitization.
Responsive UI:Mobile-friendly design using standard CSS and HTML.

2. System Architecture

The application follows a classic Model-View-Controller (MVC) inspired architecture, adapted for a PHP API + Static Frontend model.

2.1 Frontend (Client-Side)
Technology: HTML5, CSS3, Vanilla JavaScript .
Location:`public/` directory.
Communication:Interacts with the backend via asynchronous `fetch()` API calls.

Key Files:
`app.js`: Core logic, event listeners, and API wrappers.
`style.css`: Global styling and responsive layouts.
.html`: View templates.

2.2 Backend (Server-Side)
Technology: PHP, MySQL .
Location:`public/api/` directory.
Structure: RESTful API endpoints returning JSON responses.
Database Access:Uses PHP Data Objects (PDO) for secure and consistent database interactions.
Authentication:PHP Sessions (`session_start()`) based.

2.3 Files Structure

```
Electronics_Ordering_System/
└── web_project/

    └── public/
        ├── 📄 admin_dashboard.html          # Admin control panel
        ├── 📄 cart.html                     # Shopping cart page
        ├── 📄 create_test_user.php          # User management utility
        ├── 📄 customer_dashboard.html       # Customer dashboard
        ├── 📄 dashboard.html                # Order history page
        ├── 📄 index.html                    # Home page (main entry)
        ├── 📄 login.html                    # Login page
        ├── 📄 post_product.html             # Product creation/edit page
        ├── 📄 register.html                 # User registration page
        ├── 📄 seller_dashboard.html         # Seller dashboard
        ├── 📄 test_api_path.html            # API testing utility
        ├── 📄 test_login_api.php            # Login API tester
        ├── 📄 .htaccess                     # Apache configuration
        │
        ├── 📁 api/                          # Backend API endpoints
        │   ├── 📄 config.php                # Database config & helpers
        │   │
        │   ├── 📁 admin/                    # Admin-only APIs
        │   │   ├── 📄 add_stock.php         # Add inventory stock
        │   │   ├── 📄 create_user.php       # Create new user
        │   │   ├── 📄 dashboard_stats.php   # Admin dashboard statistics
        │   │   ├── 📄 delete_order.php      # Delete orders
        │   │   ├── 📄 delete_user.php       # Delete users
        │   │   ├── 📄 get_orders.php        # Get all orders
```

```
│   │   ├── 📄 get_users.php          # Get all users
│   │   └── 📄 promote_user.php       # Change user roles
│   ├── 📁 auth/                       # Authentication APIs
│   │   ├── 📄 check_session.php       # Verify user session
│   │   ├── 📄 login.php               # User login
│   │   ├── 📄 logout.php              # User logout
│   │   └── 📄 register.php            # User registration
│   ├── 📁 orders/                     # Order management APIs
│   │   ├── 📄 create.php              # Create new order
│   │   └── 📄 history.php             # Get order history
│   ├── 📁 products/                   # Product management APIs
│   │   ├── 📄 create.php              # Create new product
│   │   ├── 📄 delete.php              # Delete product
│   │   ├── 📄 get_product.php         # Get single product
│   │   ├── 📄 get_products.php        # Get all products
│   │   ├── 📄 read.php                # Read product details
│   │   └── 📄 update.php              # Update product
│   └── 📁 seller/                     # Seller-specific APIs
│       ├── 📄 all_orders.php          # Get seller's orders
│       ├── 📄 dashboard_stats.php     # Seller statistics
│       ├── 📄 notifications.php       # Seller notifications
│       └── 📄 update_order_status.php # Update order status
├── 📁 css/                            # Stylesheets
│   └── 📄 style.css                   # Main stylesheet (unified theme)
├── 📁 js/                             # JavaScript files
│   └── 📄 app.js                      # Main JavaScript functions
└── 📁 uploads/                        # Product images (auto-created)
    └── 📁 products/
```

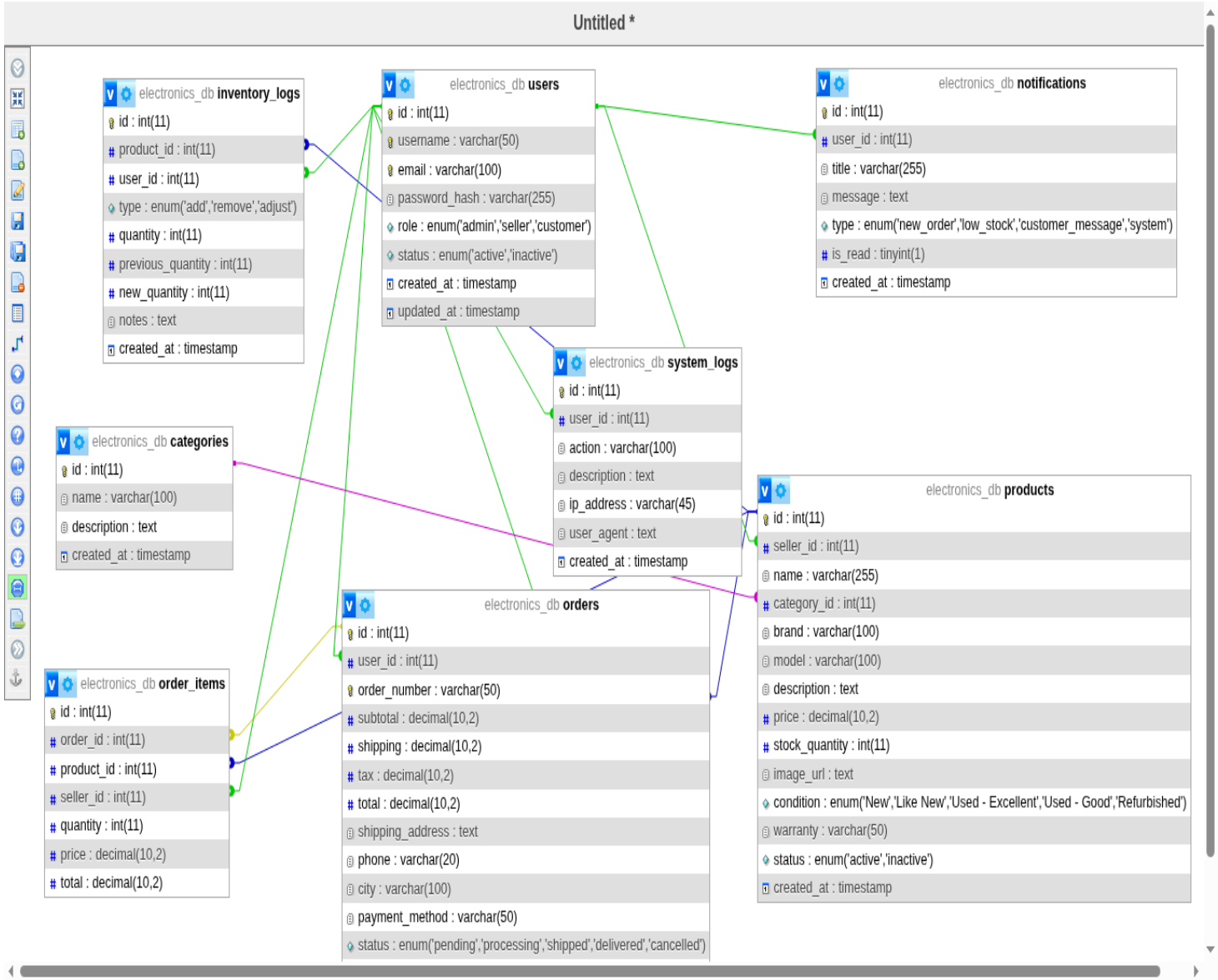
3. Database Schema

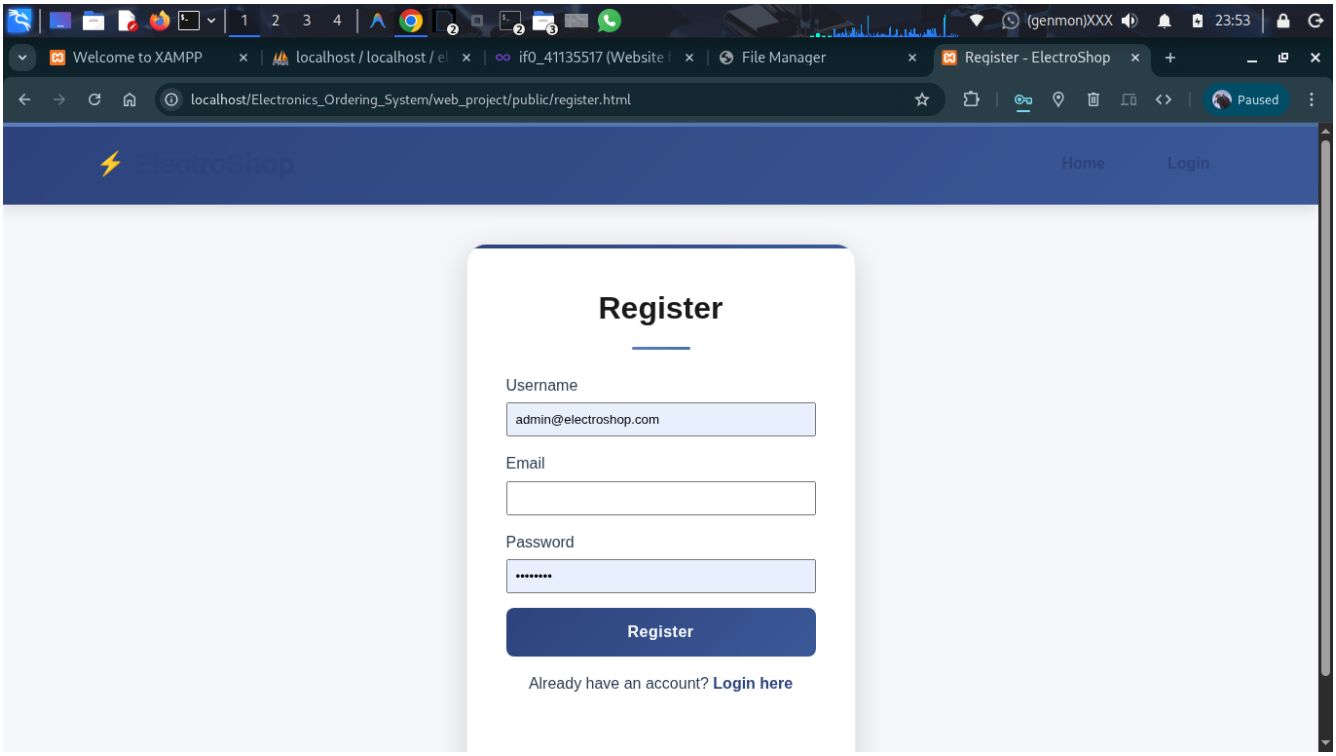
The database **`electronics_db`** consists of 9 normalized tables (3NF).

3.1 Core Tables
`users`: Stores account info.
Columns: `id`, `username`, `email`, `password_hash`, `role` (enum: 'customer', 'seller', 'admin'), `created_at`.
`products`: Inventory items.
Columns: `id`, `name`, `description`, `price`, `stock_quantity`, `category_id`, `seller_id`, `image_url`.
`categories`: Product classifications.
Columns: `id`, `name`, `description`.
`orders`: Order headers.
Columns: `id`, `user_id`, `total_amount`, `status` (enum: 'pending', 'processing', 'shipped', 'delivered', 'cancelled'), `created_at`.
`order_items`: Line items for orders.
Columns: `id`, `order_id`, `product_id`, `quantity`, `price_at_purchase`.

3.2 Support Tables

`cart`: Persistent server-side cart (optional/hybrid with local storage).
`notifications`: System alerts for users (e.g., "New Order" for sellers).
`system_logs`: Admin audit trail of critical actions.
`inventory_logs`: History of stock changes for sellers.




Welcome to XAMPP
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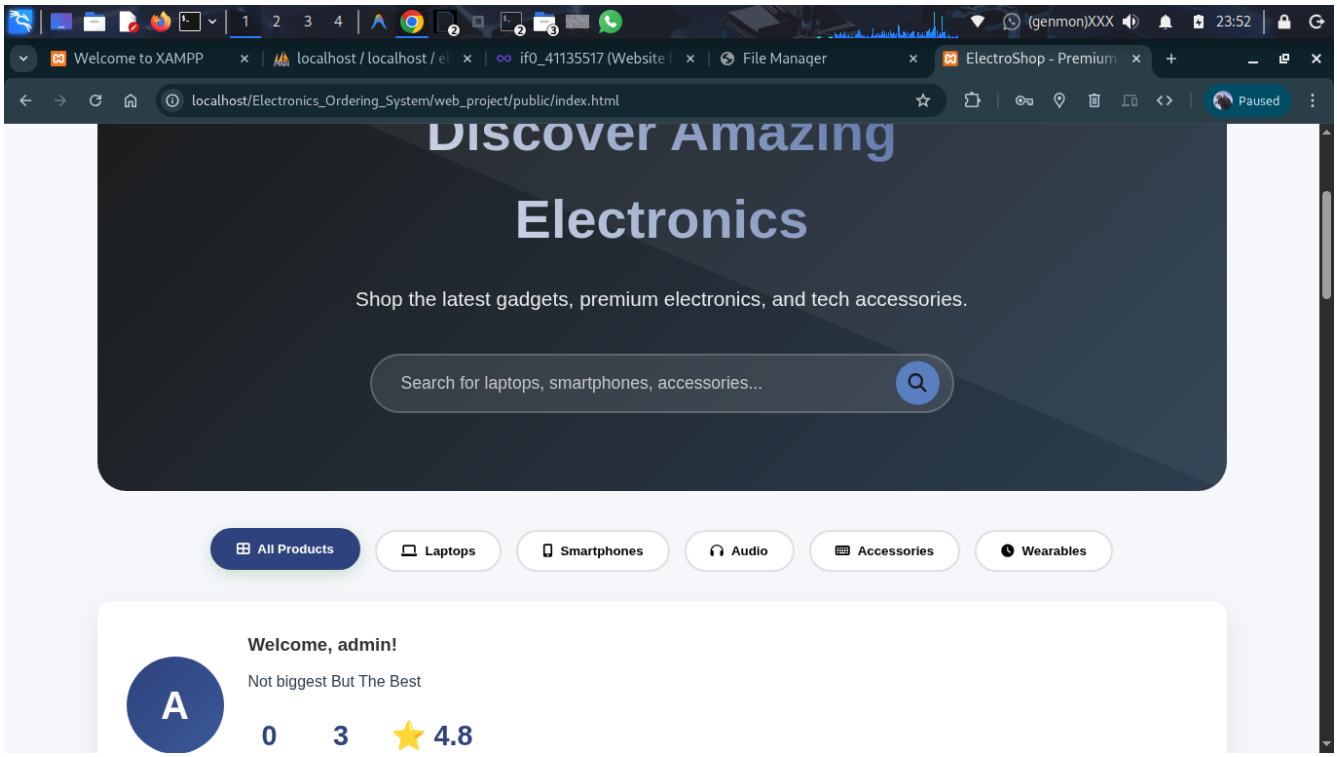
File Manager
Register - ElectroShop
localhost/Electronics_Ordering_System/web_project/public/register.html
Paused
ElectroShop
Home
Login
Register
Username
admin@electroshop.com
Email
Password
Register
Already have an account? Login here


Welcome to XAMPP
localhost / localhost / el
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File Manager
ElectroShop - Premium
localhost/Electronics_Ordering_System/web_project/public/index.html
Paused
Discover Amazing
Electronics
Shop the latest gadgets, premium electronics, and tech accessories.
Search for laptops, smartphones, accessories...
All Products
Laptops
Smartphones
Audio
Accessories
Wearables
Welcome, admin!
Not biggest But The Best
A
0
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